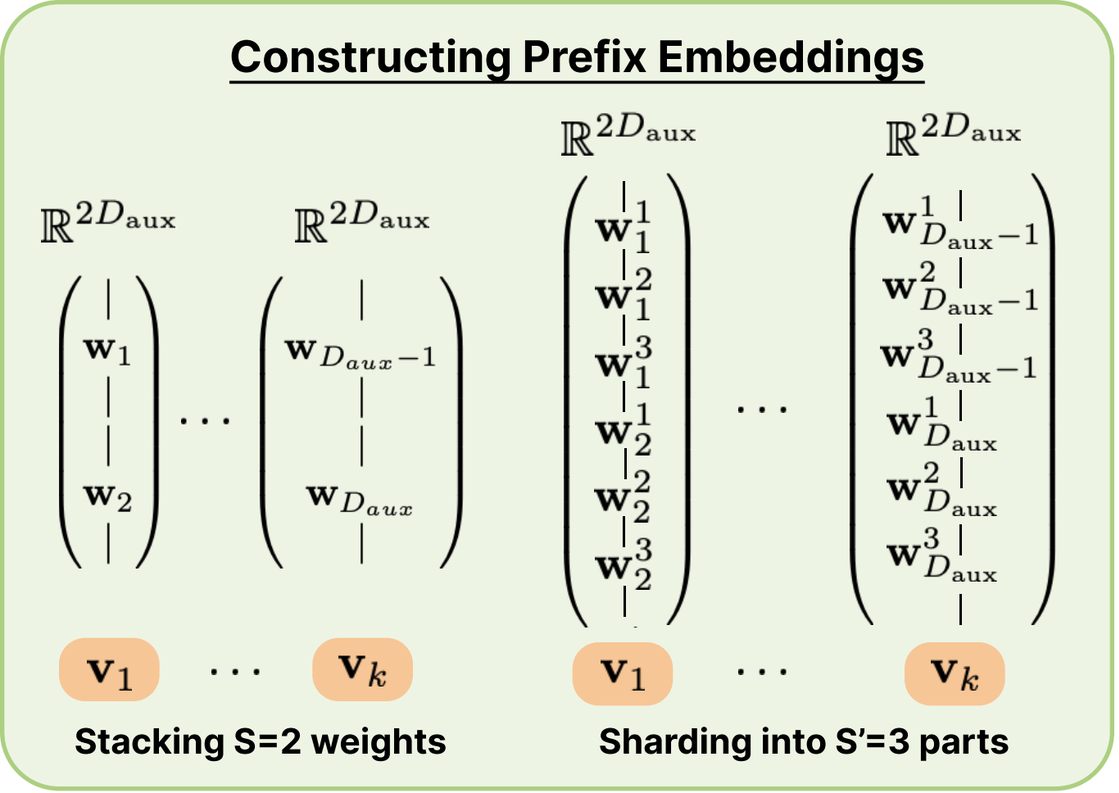
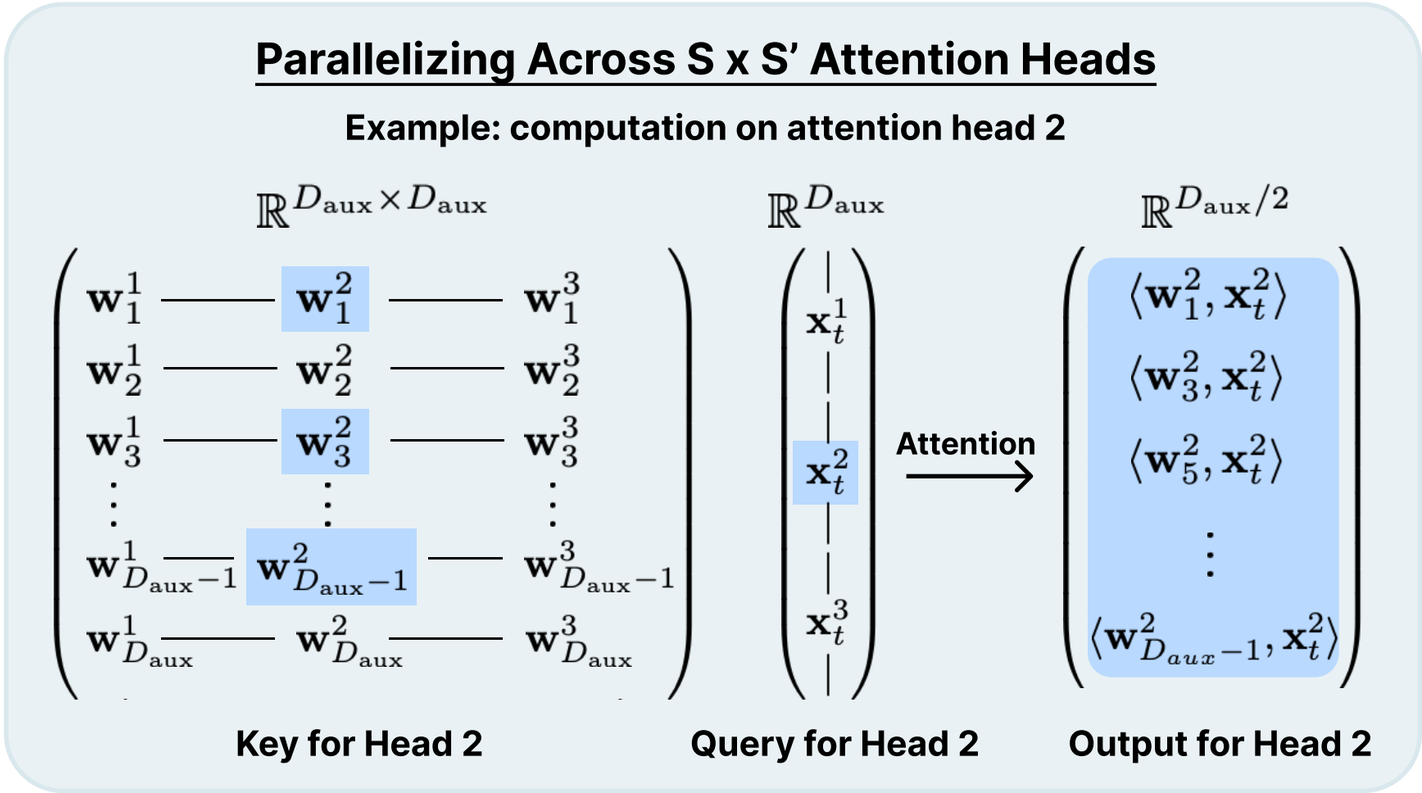

Constructing Prefix Embeddings
$\mathbb{R}^{2D_{\text{aux}}}$
$\mathbb{R}^{2D_{\text{aux}}}$
$\mathbb{R}^{2D_{\text{aux}}}$
$\mathbb{R}^{2D_{\text{aux}}}$
$\mathbf{v}_1$
$\mathbf{v}_k$
$\mathbf{v}_1$
$\mathbf{v}_k$
Stacking S=2 weights
Sharding into S'=3 parts


Parallelizing Across S x S' Attention Heads
Example: computation on attention head 2
$\mathbb{R}^{D_{\text{aux}} \times D_{\text{aux}}}$
$\mathbb{R}^{D_{\text{aux}}}$
$\mathbb{R}^{D_{\text{aux}}/2}$
Attention
Key for Head 2
Query for Head 2
Output for Head 2